

Nguyen Thach Thanh

Embedded Systems & IoT Engineering

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EDUCATION

Ho Chi Minh City University of Foreign Languages – IT (HUFLIT)

2023 – 2027

BEng in Microelectronics Engineering | Classification: Good | Thesis completed Aug 2026

PROJECTS

Buck vs. LDO Energy Characterization for a Duty-Cycled IoT Node – Undergraduate Thesis Feb 2026 – Aug 2026

- Designed a 4-layer PCB in KiCad with jumper-selectable TPS62840 buck / STLQ020 LDO branches powering an STM32L432 + BME280 node
- Wrote bare-metal register-level firmware (5 s duty cycle: BME280 forced-mode read, Stop2 sleep, RTC wake-up); one hash-verified image across all experiments
- Built an INA228 input-side energy rig on Raspberry Pi; integrated wake-to-wake energy at 15 input voltages (3.30–4.20 V) over 286 verified runs
- Found a region-dependent result: buck saves up to 17.9% per cycle at 3.8–4.2 V yet draws up to 76× more at 3.35–3.60 V — reproduced with ABBA re-runs and GPIO-marker campaigns
- **Paper:** first-author, submitted to IEEE FAIR 2026 (under review)
- **Tools:** KiCad, bare-metal C, Python (NumPy/Matplotlib), INA228, LaTeX

GreenEco – IoT Greenhouse Control System on Raspberry Pi

2025

- Built a Flask REST API driving pump/fan/light relays over active-low GPIO, with status/toggle endpoints consumed by a live dashboard
- Wrote UART/I2C sensor drivers (SEN0501 environment, SEN0220 CO₂, ES-Soil7 soil) with logging, JSON/CSV export and cloud image upload
- Unit-tested the GPIO and sensor layers
- **Tools:** Python, Flask, Raspberry Pi, GPIO, UART

AWARDS

Top 20 – Innovation Competition, Ho Chi Minh City University of Technology (HCMUT), for the GreenEco project

Top 12 – Green Innovation Competition, Hanoi University of Science and Technology (HUST), for the GreenEco project

TECHNICAL SKILLS

Embedded: C (bare-metal STM32, register-level), low-power design (Stop2, RTC wake-up, duty-cycling), I2C/UART

PCB: KiCad — 4-layer layout, buck/LDO supply design, fabrication and bench bring-up

Measurement: INA228 20-bit sensing, Python (NumPy/Matplotlib), experiment design (ABBA, marker campaigns), data reproducibility

Tools: Git, Linux (WSL), Raspberry Pi, Flask, LaTeX

LANGUAGES

Vietnamese (native) | English — working proficiency